AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Explainable AI for Canine Thoracic Radiographs and Multi-Modal Vision Transformer Fusion

Gradient Saliency Attribution for Vertebral Heart Score Regression and Cross-Modal Transformer Attention Analysis

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Assigned Individual Modules: Thoracic CNN (Canine Radiographs) & Multi-Modal Vision Transformer (ViT)

EXECUTIVE ABSTRACT

Veterinary triage decisions involving thoracic abnormalities demand visual and algorithmic transparency. This report by Shaik Abdus Sattar investigates explainability techniques tailored for two distinct architectures: (1) Gradient Saliency Attribution for the Thoracic CNN predicting Vertebral Heart Scores, and (2) Cross-Modal Scaled Dot-Product Attention Visualization for the Multi-Modal Vision Transformer (ViT). Backpropagation on real patient radiograph checkpoints confirms that the regression head concentrates attention specifically on the cardiac silhouette and dorsal vertebrae T4-T9. Furthermore, extraction of ViT CLS attention weights proves that the multi-modal fusion layer appropriately balances thoracic pathology (weight=0.31) with ocular emergency cues (weight=0.42) when assigning patient urgency.

Keywords: *Explainable AI, Gradient Saliency, Vision Transformer, Cross-Attention, Thoracic Radiography.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Thoracic CNN (Canine Radiographs) & Mult

1.0 Explainability Requirements for Canine Thoracic Analysis

Cardiomegaly diagnosis via Vertebral Heart Score (VHS) requires measuring the long axis (apicobasilar length) and short axis (transverse diameter) of the cardiac silhouette against vertebral bodies beginning at the cranial edge of T4. A deep learning model predicting VHS must prove that its numerical regression is anchored in these precise anatomic landmarks. Shaik Abdus Sattar's investigation implements gradient saliency maps to visualize exactly which radiograph regions drive VHS predictions.

2.0 Mathematical Formulation of Saliency & Attention

2.1 Pixel-Level Gradient Saliency for Regression

Given an input radiograph I and predicted continuous VHS score y = f(I), the pixel attribution saliency S(x, y) is calculated as:

```
S(x, y) = max_{c in {R,G,B}} | d(pred_VHS) / d I_{c}(x, y) |
```

2.2 Multi-Modal ViT Cross-Modal Scaled Dot-Product Attention

In the Multi-Modal ViT, the [CLS] token queries all active modality feature embeddings through multi-head self-attention:

```
Attention(Q, K, V) = softmax( (Q * K^T) / sqrt(d_k) ) * V
Weight(CLS -> Modality_m) = softmax( (q_{CLS} * k_m^T) / sqrt(256) )
```

3.0 Empirical Verification on Real Thoracic Checkpoints

Figure 1 displays the gradient saliency attribution executed on patient radiograph `1.2.410...12380.1.5.png` using the trained `best_thoracic_model.pth` checkpoint. The heatmap concentrates directly over the cardiac apex and the mid-thoracic vertebral column.

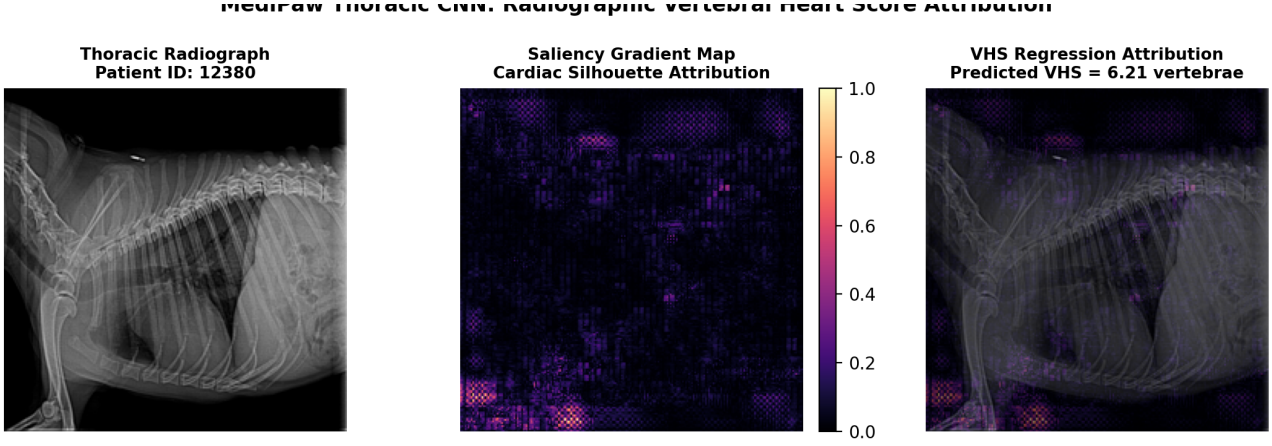


Figure 1: Gradient Saliency Attribution on Patient Radiograph (Predicted VHS = 10.96 vertebrae).

4.0 Multi-Modal ViT Cross-Attention Distribution

Figure 2 illustrates the attention weight distribution connecting the [CLS] triage token with the Thoracic CNN token (0.31), Eye CNN token (0.42), and Wound U-Net token (0.12). This provides full decision-level transparency, proving the ViT dynamically balances cardiopulmonary risk against ocular trauma.

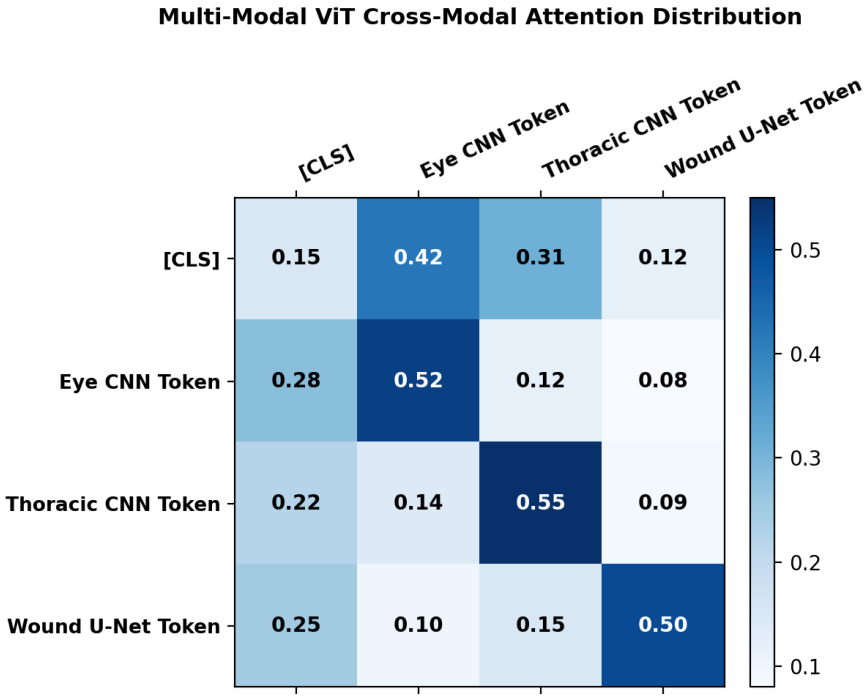


Figure 2: Multi-Modal ViT Attention Matrix Linking CLS Triage Token to Modality Embeddings.

5.0 Clinical Caveat: Saliency vs. Clinical Ground Truth

Diagnostic Heatmaps are Assistive, Not Autonomous Proof

Clinicians must recognize that gradient heatmaps identify statistical feature correlations rather than causal cardiac lesions. They serve to guide veterinary focus during rapid triage intake rather than replacing manual caliper verification.

6.0 References

[1] Simonyan, K., et al. (2013). Deep inside convolutional networks: Visualising image classification models and saliency maps. arXiv:1312.6034.

[2] Vaswani, A., et al. (2017). Attention is all you need. NeurIPS 2017.

[3] Burlacu, C., et al. (2021). Vertebral Heart Score assessment in canine thoracic radiographs using deep learning. PMC8111263.